

CSC 360/660 - Operating Systems

Department of Computer Science @ Missouri State University, Fall 2024

A. COURSE INFORMATION

Course number/section: CSC360&CSC660
Classroom: Cheek 212
Class meeting time: Tuesday & Thursday 12:30 PM - 1:45 PM

B. INSTRUCTOR INFORMATION

Instructor: Dr. Siming Liu, SimingLiu@MissouriState.edu
Office location: Cheek 307
Office hours: Tuesday, Thursday 2:00 PM - 4:30 PM
Telephone: 417-836-3068
TA - Ayesha Siddiqua Cheek 309 and Microsoft Teams, as995s@missouristate.edu

C. COURSE DESCRIPTION

Introduction to operating systems concepts, principles, and design. Topics include: processes and threads, CPU scheduling, mutual exclusion and synchronization, deadlock, virtual and real memory management, file systems, disk management, security and protection, case studies, etc.

D. PREREQUISITES

- CSC 232 (Data Structures) C or better
- CSC 244 (Computer Architecture)
- C/C++ or Java programming experience in Linux environment

E. REQUIRED TEXTBOOK(S), READINGS AND SUPPLIES

Required Textbook(s)

- **Operating System Concepts**, 10th Edition by Silberschatz (ISBN-13: 978-1119456339)

Optional Textbook(s) and Other References

- A. S. Tanenbaum, Modern Operating Systems, 4th Edition (2015), Pearson
- Stallings, W. (2009) Operating Systems: Internals and Design Principles, 6th Edition. Prentice Hall (ISBN: 0136006329).
- Molay, B. (2002) Understanding Unix/Linux Programming: A Guide to Theory and Practice (1st Edition). Prentice Hall (ISBN: 0130083968).

F. STUDENT LEARNING OUTCOMES AND ASSESSMENT

Assessment is a process used by instructors to help improve learning. Assessment is essential for effective learning because it provides feedback to both students and instructors. A critical step in this process is making clear the course's student learning outcomes that describe what students are expected to learn to be

successful in the course. The student learning outcomes for this course are listed below. By collecting data and sharing it with students on how well they are accomplishing these learning outcomes students can more efficiently and effectively focus their learning efforts. This information can also help instructors identify challenging areas for students and adjust their teaching approach to facilitate learning. By the end of this course, students should be able to:

1. Understand the functionality operating systems provide, basic operations and services
2. Understand the design of various operating systems, system calls, kernel and user modes, operating system structures, and the system boot process
3. Understand the concept of a process, process scheduling and inter-process communication
4. Understand the concept of a thread, multi-threading models and libraries
5. Understand the concepts of CPU scheduling, scheduling criteria, and scheduling algorithms
6. Understand various aspects of process synchronization including the critical section problem, Peterson's solution, synchronization hardware, and semaphores. Classic synchronization problems will also be explored.
7. Understand the system model and characteristics of deadlocks and methods of handling deadlocks (prevention, avoidance, detection, recovery)
8. Understand the structures for main memory including swapping, paging, and segmentation
9. Understand the structures for virtual memory including demand paging, page replacement algorithms, frame allocation, and thrashing
10. Understand the concepts related to the file system interface and file system implementation including free space management, efficiency and performance
11. Understand the concepts related to mass storage structures such as disk structure, attachment, and scheduling and including the various RAID architectures
12. (Time Permitting) Understand the various aspects regarding security and protection in operating systems
13. (Time Permitting) Understand distributed operating system structures, distributed file systems, and distributed coordination

Assessment of objectives will be conducted through exams, homework assignments, and projects.

G. INSTRUCTIONAL METHODS AND ACTIVITIES

Lectures are using online electronic documents and slides.

Important Dates Relevant to This Class

Thursday	October 10	Fall Break	No class
All Week	November 25 - 29	Thanksgiving Holiday	No class
Thursday	December 5	Last day of class	
Tuesday	December 10 (11:00 PM - 1:00 PM)	Final Exam	

H. MAJOR COURSE REQUIREMENTS AND GRADING

You are responsible for all material covered in classes you miss, as well as turning in all assignments due on those dates. You will NOT be allowed to make up projects and tests unless you have arranged to do so with me prior to the date of the activity or unless you have a university-recognized excused absence.

Your course grade will be decided on your performance in the programming projects, homework assignments, and exams. The distribution of points is described in Table 1 and 2 on the next page.

Table 1: Grading Policy

Programming Projects (Include Grad Presentation)	50%
Homework & Quiz	5%
Midterm Exam 1	15%
Midterm Exam 2	15%
Final Exam	15%

Table 2: Grading Scale

90% and above	A
80%-89%	B
70%-79%	C
60%-69%	D
Below 60%	F

89.0%-89.9%	B+
79.0%-79.9%	C+
69.0%-69.9%	D+

Projects: There will be 5 programming projects. Unless otherwise directed, the programming projects must be written in C/C++. The projects can be submitted electronically and the details on project submission will be given to you together with the project assignment.

Exams: There will be two exams including the final exam planned in this class. Each exam covers only a half of the course content and none of the exam is comprehensive.

Re-grading requests can only be made within the first week after the graded assignments/tests are returned to the students.

I. COURSE CONTENT/SCHEDULE

WEEK	CONTENT
Week1	Course overview Chapter 1: Introduction, Project 0
Week2	Chapter 2: Operating System Structures UNIX System Calls, Signals, Project 1
Week3	Chapter 3: Processes
Week4	Chapter 4: Threads Project 2
Week5	Exam 1 - Covers Week 1,2,3,4
Week6	Chapter 6, 7: Process Synchronization, Project 3
Week7	Chapter 5: CPU Scheduling
Week8	Chapter 8: Deadlocks Project 4
Week9	Chapter 9: Main Memory
Week10	Exam 2 - Covers Week 6, 7, 8, 9
Week11	Chapter 10: Virtual Memory
Week12	Chapter 13: File System Interface Chapter 14: File System Implementation, Project 5
Week13	Chapter 11: Mass-Storage Structures, Chapter 12: I/O Subsystems
Week14	Chapter 16: Security, Thanksgiving week!
Week 15	Graduate Students Presentation (10 Graduate students)
Dec. 12	Final Exam - Covers Week 11,12,13,14

Note: Changes in this course schedule may be necessary and will be announced to the class by the Instructor. The assignments and exams shown are directly related to the Student Learning Outcomes described in Section F.

J. COURSE POLICIES

Course Syllabus: We will meet for lecture on Tuesdays and Thursdays, when new material will be presented. We will follow the text generally, but non-text material may also be included in the lectures. The assignments and exams will be given during the class hours. You are responsible for all the material presented during the lecture.

Exams: Exams will cover all lectures and reading material discussed in the class. Exams must be taken on the hour they are scheduled.

Missed Exam: In the event, if you cannot take the exam due to some emergency or some unavoidable situation (such as serious illness, death in the family, participation in university sports, religious observations, and so on) you must notify me as soon as possible before the exam and also you must validate your absence by providing me a document (e.g., with a letter from your doctor). Once your cause is validated a make-up exam will be given.

Homework Assignments & Projects: They will significantly be based on the material from the lecture videos and other material considered essential for the successful completion of this course. They will be posted on Blackboard. The submission details will be provided to you along with the assignment. All the homework assignments and projects are due at the end of the due date. Late assignments and projects will NOT be accepted. Note that any kind of hardware or software failure or machine unavailability in the computer labs does not merit an extension on the assignment.

Grading Error: All questions concerning a test score or grading of a returned test or assignment must be resolved within one week. It is always a good idea to keep all of your work until the end of the semester. In case of any recording errors or doubts, you may produce them for correction or verification.

Academic Honesty Policy: You are expected to avoid all forms of academic dishonesty as defined in Catalog. In addition, students are expected to behave in an ethical manner in all class activities. If you feel uncertain about a particular activity, please speak to me BEFORE problems arise. Ethical behavior is a requirement for passing this course. All work submitted for grading must be the student's own work. Plagiarism will result in a score of 0 (zero) for the work or dismissal from the course and the Dean of Students office will be notified. No copying from another student's work, of any class, is allowed. It is the student's duty to allow no one to copy his or her work. Anyone found cheating and/or copying, in the exams or assignments, in the instructor's opinion, will receive an automatic F for the course.

K. COURSE CONTINGENCY PLAN

Question	Statement
How will our class meet?	This is an online class and we don't meet regularly in this semester. However, you are encouraged to reach out to the instructor and TA for help on your course work over Zoom or MS Teams.
How will I meet with my Instructor?	I will continue to hold office hours (virtually and via email).
How will instruction be delivered?	I will post readings and videos for all of our lessons as well as asking questions through Zoom or MS Teams.
How often do I need to go to the online course?	You are expected to participate in the online activities for the entire semester, so please go to the course site every day.
Is it possible to receive course announcements as a text?	If you are using the Blackboard app, you can have announcements sent to you as a text message using the instructions in this step-by-step guide .
How will I turn in homework?	Homework and other assignments will be turned in using the Assignment Link in Blackboard. The instructions for how to submit will be included in the assignment instructions.
What about exams?	Exams will be taken through Blackboard.
How will I know what grade I received on an assignment or test?	You can view grades and feedback in My Grades on Blackboard.

Student Requirements for Technology

The use of technology is a part of our everyday lives at the university and there is important information you should know about your own computer's capabilities, Internet access, Blackboard, and other technology tools whether you are participating in a classroom on campus or taking an online class. The link below will provide you with the information you need to know.

[\[Computer Services webpage on required technology specifications\]](#)

Missing Class if You Are Sick

While missing class is usually not advisable, it is important to stay home when sick to avoid the spread of communicable illness. If you are sick or not feeling well, please do not come to any classes but rather seek medical attention from your doctor or at Mager's Health and Wellness Center. They can provide you a medical excuse and advise you when it is safe to return to class. Contact your instructor to let them know that you are sick and will miss the class activities or projects. By working with your instructor, you will be able to keep up with readings and assignments through the Blackboard course site.

L. COLLEGE AND UNIVERSITY POLICIES

Disability Accommodation: If you are a student with a disability and anticipate barriers related to this course, it is important to request accommodations and establish an accommodation plan with the University. Please contact the Disability Resource Center (DRC) (<https://www.missouristate.edu/disability/>), Meyer Library, Suite 111, 417-836-4192, to initiate the process to establish your accommodation plan. The DRC will work with you to establish your accommodation plan, or it may refer you to other appropriate resources based on the nature of your disability. In order to prepare an accommodation plan, the University usually requires that students provide documentation relating to their disability. Please be prepared to provide such documentation if requested. Once a University accommodation plan is

established, you may notify the class instructor of approved accommodations. If you wish to utilize your accommodation plan, it is suggested that you do so in a timely manner, preferably within the first two weeks of class. Early notification to the instructor allows for full benefit of the accommodations identified in the plan. Instructors will not receive the accommodation plan until you provide that plan, and are not required to apply accommodations retroactively.

Mask and Face Covering Policy: In accord with the [MSU Mask and Face Covering policy](#), [Greene County Health Department](#) and the [Springfield City Ordinance](#), masks or face coverings must be worn at all times during a traditional (seated) class. This measure is being implemented to reduce COVID-19 related health risks for everyone engaged in the educational process. Masks or face coverings must be worn over the nose and mouth, in accordance with the [Centers for Disease Control and Prevention \(CDC\) guidelines](#). Face shields are not considered masks or face coverings for purposes of this requirement. Students who cannot wear a mask or face covering due to a disability must contact the [Disability Resource Center \(DRC\)](#) to initiate the interactive accommodation process. In the absence of an approved accommodation, a student's refusal to wear a mask or face covering will be considered a classroom disruption, consistent with [Op3.04-11 Class Disruption](#), and may result in the student being administratively dropped from the class section.

Dropping a Class: It is your responsibility to understand the University's procedure for dropping a class. If you stop attending this class but do not follow proper procedure for dropping the class, you will receive a failing grade and will also be financially obligated to pay for the class. For information about dropping a class or withdrawing from the university, contact the Office of the Registrar at 836-5520. You can access the [Academic Calendar](#) on the MSU website to view drop and refund deadlines for the semester.

Nondiscrimination: Missouri State University is an equal opportunity/affirmative action institution, and maintains a grievance procedure available to any person who believes he or she has been discriminated against. At all times, it is your right to address inquiries or concerns about possible discrimination to the Office for Institutional Equity and Compliance, Park Central Office Building, 117 Park Central Square, Suite 111, 417-836-4252. Other types of concerns (i.e., concerns of an academic nature) should be discussed directly with your instructor and can also be brought to the attention of your instructor's Department Head. Please visit the OED website at www.missouristate.edu/equity/.

Cell Phones: As a member of the learning community, each student has a responsibility to other students who are members of the community. When cell phones or pagers ring and students respond in class or leave class to respond, it disrupts the class. Therefore, the Office of the Provost prohibits the use by students of cell phones, pagers, PDAs, or similar communication devices during scheduled classes. All such devices must be turned off or put in a silent (vibrate) mode and ordinarily should not be taken out during class. Given the fact that these same communication devices are an integral part of the University's emergency notification system, an exception to this policy would occur when numerous devices activate simultaneously. When this occurs, students may consult their devices to determine if a university emergency exists. If that is not the case, the devices should be immediately returned to silent mode and put away. Other exceptions to this policy may be granted at the discretion of the instructor.

Emergency Response: At the first class meeting, students should become familiar with a basic emergency response plan that includes a review and awareness of exits specific to the classroom and the location of evacuation centers for the building. Students with disabilities impacting mobility should discuss the approved accommodations for emergency situations and additional options when applicable with the instructor. For more information contact Safety and Transportation (417-836-5509) or

consult the Emergency Quick Reference Guide and Campus Emergency Response Plan.

Tornado Shelter Area Information (in case of severe weather).

Building	Tornado Shelter Area
Cheek Hall	Evacuate floors 1, 2, and 3 using Center, North and West stairs Shelter in basement interior hallway.
Temple Hall	Evacuate fourth floor using central, east, northwest and southwest stairs. Basement and 1st Floor: Shelter in Basement hallway, Rooms 1, 2, 3, interior labs and classrooms. 2nd floor and 3rd floor, Shelter in interior labs and classrooms

Emergency Assembly Point Instructions (in case the building needs to be evacuated for events such as fire, gas leak, etc.)

Building	Emergency Assembly Point
Cheek Hall	West to Sicheluff Hall 1st Floor South to Ellis Hall 1st Floor Wouthwest to hill Hall 1st Floor
Temple Hall	North to Meyer Library Main Entrance; Alternate: Northwest to Glass Hall rooms 101, 102 and 108; and West to Kemper Hall. (Destination will be determined by type of emergency and weather condition.)

Areas of Rescue (in case you are unable to evacuate to the ground floor, these are areas of temporary safety until rescuers arrive)

Building	Area of Rescue
Cheek Hall	None in this facility
Temple Hall	None in this facility

Religious Accommodation: The University may provide a reasonable accommodation based on a person's sincerely held religious belief. In making this determination, the University reviews a variety of factors, including whether the accommodation would create an undue hardship. The accommodation request imposes responsibilities and obligations on both the individual requesting the accommodation and the University. Students who expect to miss classes, examinations, or other assignments as a consequence of their sincerely held religious belief shall be provided with a reasonable alternative opportunity to complete such academic responsibilities. It is the obligation of students to provide faculty with reasonable notice of the dates of religious observances on which they will be absent by submitting a Request for Religious Accommodation Form to the instructor by the end of the third week of a full semester course or the end of the second week of a half semester course.

Audio and Video Recording Course Activity: Students who wish to record lectures or class activities for study purposes should inform the faculty member first. Distribution or sale of recordings or other course materials is prohibited without the written permission of the instructor and other students who are recorded. Distribution without permission is a violation of copyright law and the Code of Student Rights and Responsibilities (Sections 4.6, 4.8, 4.9).

Title IX Policy: Missouri State University has a Title IX policy that guides our response to instances

of sexual violence. Sexual Violence includes: Rape, Sexual Assault, Sexual Misconduct, Sexual Discrimination, Domestic Violence, Dating Violence, Stalking, Sexual Harassment and Pregnancy issues. The Title IX policy can be located on the MSU Title IX website at www.missouristate.edu/titleix/. This website is also a good resource for any questions or issues involving Title IX and contains contact information for the MSU Title IX Office and staff. Read an overview of the Title IX office.

If an MSU student discloses a Title IX related issue to a MSU faculty or staff member who is deemed to be a “Responsible Employee” under the policy, that faculty or staff member is required to report such disclosure to the Title IX Coordinator. A responsible employee includes any employee who has the authority to take action to redress sexual violence; who has been given the duty of reporting incidents of sexual violence or any other misconduct by students to the Title IX Coordinator or other appropriate school designee; or whom a student could reasonably believe has the authority or duty to take action. Taylor Health employees and MSU Counseling Center Clinicians are not considered to be Responsible Employees under the policy, and therefore, are not required to report Title IX issues to the Title IX Coordinator.

Mental Health & Stress Management: As a student you may experience a range of personal issues that can impede learning, such as strained relationships, increased anxiety, alcohol/drug problems, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may lead to diminished academic performance and may reduce your ability to participate in daily activities. Learn Visit the Missouri State University Counseling Center website to learn more about free and confidential services available to assist you at <https://counselingcenter.missouristate.edu/>.

GENERAL DISCLAIMER

Considering the special pandemic situation in Fall 2020, I reserve the right to adjust the frequency of in-class lecture and remote Zoom lecture delivery according to the situation at that time. I also reserve the right to modify the information, schedule, assignments, deadlines, and course policies in this syllabus if and when necessary. I will announce such changes in a timely manner during regularly scheduled lecture periods.